INDIVIDUALS POST-STROKE CHANGE ENERGETICS OF MULTIPLE LOWER LIMB CONSTITUENTS DURING GAIT WHILE WEARING A PASSIVE-DYNAMIC ANKLE FOOT ORTHOSIS

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Introduction

Passive-dynamic ankle-foot orthoses (PD-AFOs) are commonly prescribed to individuals post-stroke to supplement weakened plantar-flexor muscles in their paretic limb. Our previous research demonstrated stiffness-customized PD-AFOs capacity to reduce lower extremity mechanical cost of transport (COT) for individuals post-stroke when walking compared to walking with No AFO or with their standard-of-care (SOC) AFO. [1] The purpose of this study was to investigate how individuals achieve this COT reduction.

We hypothesized COT reductions would be driven primarily by decreases in the subjects' paretic ankle work.

Methods

Eight individuals (mean (SD) age 65.1 (6.9) years; height 1.72 (0.13) m; mass 82.4 (14.8) kg; 3 male, 4 female) with single chronic stroke took part in this study. Instrumented gait analysis was conducted using an instrumented treadmill and motion capture cameras as participants walked without any AFO and then with their SOC AFO at their self-selected waking speed determined via a 10 Meter Walk Test without an AFO. A PD-AFO was stiffness customized and manufactured for each participant based on the peak paretic plantar-flexor moment during stance from the No AFO data. Participants were given the PD-AFO to wear for four weeks, then returned to the lab for an instrumented gait analysis with the PD-AFO at the same self-selected walking speed from the initial visit.

Data were processed and analyzed in Visual 3D. COT was calculated per limb for each condition as positive limb work (summed hip, knee, ankle, and distal foot, all normalized by body mass) summed with the absolute value of negative limb work over the gait cycle, scaled by stride length. [1] Finally, The Constituent Lower Extremity Work (CLEW) approach was used to quantify and visualize how energetics of the 14 lower extremity constituents (bilateral positive and negative ankle, knee, hip during stance and swing, and positive and negative distal foot during stance only) changed with PD-AFO use. [2]

Results and Discussion

All subjects with full data sets saw decreases in total COT (paretic + nonparetic limbs) with the PD-AFO compared to No AFO and all but one decreased total COT compared to the SOC AFO. While wearing the PD-AFO, six subjects decreased the net COT in their paretic limb and seven decreased the net COT in their nonparetic limb compared to No AFO. Conversely, seven decreased the net COT in their paretic limb and six decreased the net COT in their nonparetic limb compared to their SOC AFO (Table 1). Interestingly, on average, subjects showed reductions in over half of the constituents' work with the PD-AFO use compared to both the No AFO and SOC AFO conditions (Table 1), and these changes were all relatively small (<0.19 J/kg).

The results of this study support our previous findings that showed stiffness-customized PD-AFOs enable COT reductions. However, the results do not support our hypothesis as to how these reductions were achieved. Instead of total COT reductions being driven by the ankle or even a few primary constituents, results indicated that the decrease in total COT was due to a summation of small changes across the entirety of both limbs. This suggests that, for individuals post-stroke, PD-AFOs do not simply cause direct enhancement of plantar flexor function; instead an adaptation or reorganization of much of the lower limb system appears to occur.

Table 1. Number of lower limb constituents that decreased work magnitude and magnitude of change in COT (J/kg/m) with the PD-AFO compared to both the No AFO and SOC AFO. ±COT could not be calculated due to technical error.

	No AFO vs PD-AFO				**SOC AFO vs PD-AFO**			
	Paretic		*Nonparetic*		*Paretic*		*Nonparetic*	
Subject	*#*	*ΔCOT*	*#*	*ΔCOT*	*#*	*ΔCOT*	*#*	*ΔCOT*
P1	5	-0.01	11	-0.13	3	-0.01	6	-0.1
P2	4	-0.05	11	-0.29	5	0.12	8	-0.31
P3	12±	-±	-±	-±	11	-0.55	12	-0.16
P4	10	0	13	-0.2	9	-0.06	7	0.02
P5	14	-0.04	12	-0.54	14	-0.05	10	-0.17
P6	8	-0.55	7	-0.4	9	-0.32	6	-0.58
P7	7	-0.54	5	-0.17	4	-0.03	5	0.19
P8	8	-0.27	12	-0.04	9	-0.59	9	-0.01
Mean	*8.5*	*-0.21*	*8*	*-0.25*	*10*	*-0.19*	*8*	*-0.14*
SD	*3.38*	*0.25*	*3.74*	*0.17*	*2.97*	*0.27*	*2.36*	*0.23*

Significance

This study provided valuable insight into how the post-stroke musculoskeletal system interacts with stiffness-customized PD-AFOs. Further, this study highlighted the importance of looking at the entirety of the lower limbs rather than just the constituent being targeted by a given assistive device. Future work can investigate whether the change in COT observed while wearing a stiffness-customized PD-AFO corresponds to a change in muscle activation patterns. Additionally, future studies can investigate whether subjects' self-organized changes are optimal for reducing COT or if training can improve their performance. This study combined with future work will help to optimize the design and prescription of PD-AFOs as well as rehabilitation and training strategies to meet patients' needs and improve their quality of life.

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References

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